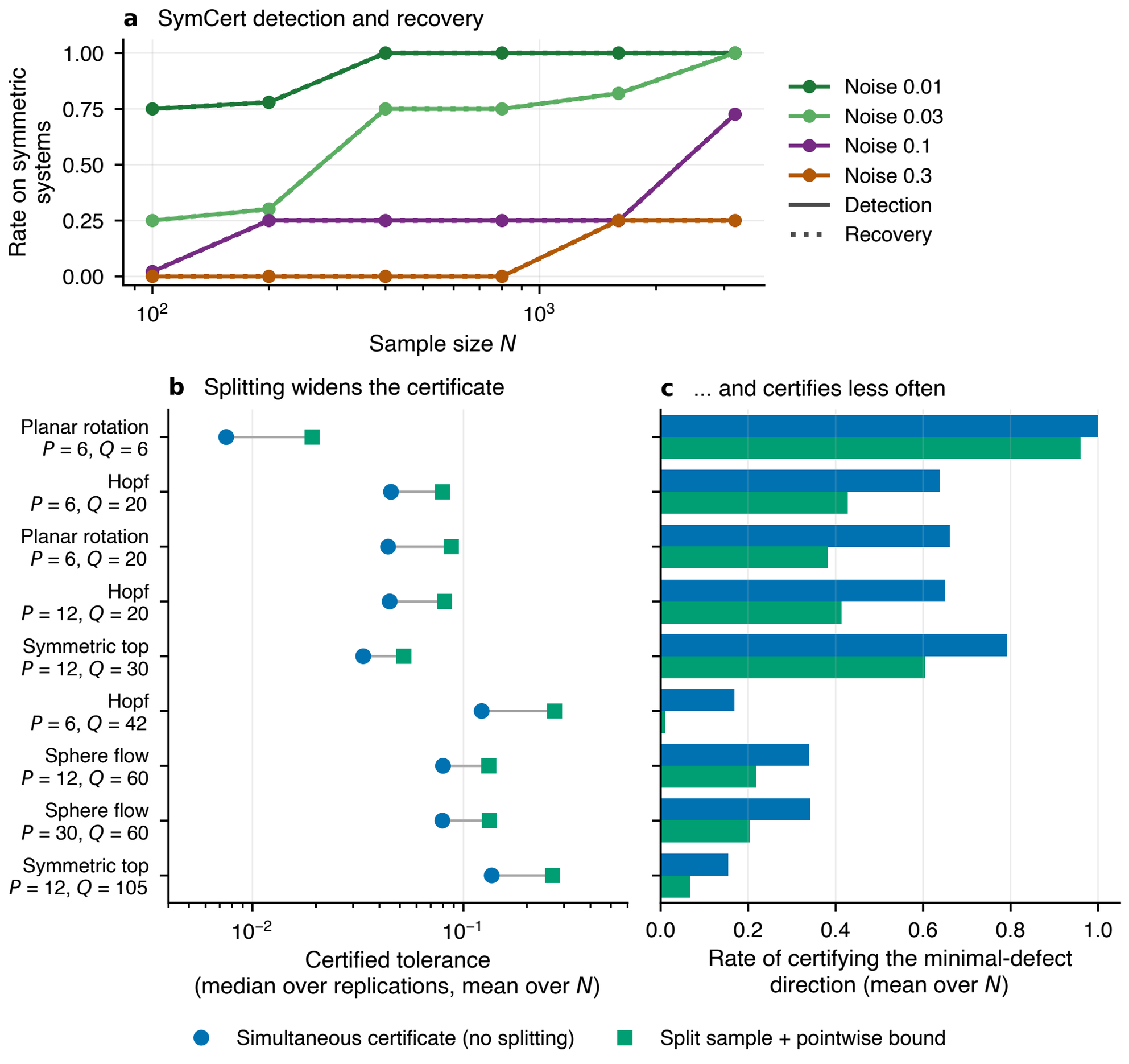
a SymCert detection and recovery
Rate on symmetric systems
1.00
0.75
0.50
0.25
0.00
10^2
10^3
Sample size N
Noise 0.01
Noise 0.03
Noise 0.1
Noise 0.3
Detection
Recovery
b Splitting widens the certificate
Planar rotation $P = 6, Q = 6$
Hopf $P = 6, Q = 20$
Planar rotation $P = 6, Q = 20$
Hopf $P = 12, Q = 20$
Symmetric top $P = 12, Q = 30$
Hopf $P = 6, Q = 42$
Sphere flow $P = 12, Q = 60$
Sphere flow $P = 30, Q = 60$
Symmetric top $P = 12, Q = 105$
10^{-2}
10^{-1}
Certified tolerance
(median over replications, mean over N)
c ... and certifies less often
0.0
0.2
0.4
0.6
0.8
1.0
Rate of certifying the minimal-defect direction (mean over N)
Simultaneous certificate (no splitting)
Split sample + pointwise bound